Answer Key & Brief Solutions

1. $AB = 4 - (-11) = 15$. With $AC = 7$ and A–C–B: $BC = AB - AC = 15 - 7 = 8$.

2. Internal $2:3$ from A to B: $P = \frac{3A+2B}{2+3} = \frac{3(5)+2(-10)}{5} = \frac{15-20}{5} = -1$.

3. $M = \left(\frac{-7+5}{2}, \frac{2+(-8)}{2}\right) = (-1, -3)$; $EF = \sqrt{(5+7)^2 + (-8-2)^2} = \sqrt{12^2 + (-10)^2} = \sqrt{144 + 100} = \sqrt{244} = 2\sqrt{61}$.

4. Area $= \frac{1}{2} \cdot 9 \cdot 6 = 27$. Perimeter: $GH = 9$, $HJ = \sqrt{(3-9)^2 + (6-0)^2} = \sqrt{36+36} = 6\sqrt{2}$, $JG = \sqrt{(0-3)^2 + (0-6)^2} = \sqrt{9+36} = 3\sqrt{5}$. So $P = 9 + 6\sqrt{2} + 3\sqrt{5}$.

5. $r = |8 - 2| = 6$. $C = 2\pi r = 12\pi$. $A = \pi r^2 = 36\pi$.

6. Straight line $\Rightarrow$ sum $180°$: $(4x - 6) + (2x + 24) = 180 \Rightarrow 6x + 18 = 180 \Rightarrow x = 27$. Angles: $4(27) - 6 = 102°$ and $2(27) + 24 = 78°$.

7. Same-side interior angles on parallel lines are supplementary: $(5y + 15) + (3y + 35) = 180 \Rightarrow 8y + 50 = 180 \Rightarrow y = 16.25$. Angles: $5y + 15 = 5(16.25) + 15 = 96.25°$, $3y + 35 = 83.75°$ (sum $180°$).

8. Bisector: $x + 12 = 3x - 24 \Rightarrow 2x = 36 \Rightarrow x = 18$. Then each half is $30°$, so $\angle PQS = 60°$.

9. Reflect across y-axis: $(x, y) \mapsto (-x, y)$. $A'(-4, -3)$, $B'(-1, 5)$. Translate by $\langle -2, 4 \rangle$: $A''(-6, 1)$, $B''(-3, 9)$. Reflections/translations are rigid motions, so lengths/perimeter are preserved.

10. $90°$ CW: $(x, y) \mapsto (y, -x)$, so $R(-6, 2) \mapsto (2, 6)$. Then $180°$: $(x, y) \mapsto (-x, -y)$, so $(2, 6) \mapsto (-2, -6)$.

11. $N(2, 7)$. Area $= |KL| \cdot |LM| = 7 \cdot 6 = 42$. Diagonal $KN = \sqrt{(2-2)^2 + (7-1)^2} = 6$.

12. Hypotenuse $= \sqrt{(x+1)^2 + x^2} = \sqrt{2x^2 + 2x + 1}$. For $x = 7$: $\sqrt{98 + 14 + 1} = \sqrt{113}$ (not integer) $\Rightarrow$ not a Pythagorean triple.

13. Strip area $=$ (circumference)$\times$height $= 2\pi r \cdot h = 180$ with $h = 6$. So $2\pi r \cdot 6 = 180 \Rightarrow r = \dfrac{180}{12\pi} = \dfrac{15}{\pi}$ cm; height $= 6$ cm.

14. $s = \sqrt{r^2 + h^2} = \sqrt{25 + 144} = 13$ cm. Total $SA = \pi r(r + s) = \pi \cdot 5(5 + 13) = 90\pi$ cm^2. Volume $V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi \cdot 25 \cdot 12 = 100\pi$ cm^3.

15. $\dfrac{4}{3}\pi r^3 = \dfrac{2048}{3}\pi \Rightarrow r^3 = 512 \Rightarrow r = 8$ cm. $SA = 4\pi r^2 = 4\pi \cdot 64 = 256\pi$ cm^2.